MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE) TYPE 5893

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: UHF pencil triode, glass-metal

Mounting position: Any

See figure 1

Weight: 0.5 ounce nominal

ABSOLUTE RATINGS: MAXIMUM

Parameter:	F1	Ef	Eb	Ec	eb	Ib	ib	ec	Ic
Unit:	MHz	V	Vdc	Vdc	v	mAdc	a	Vac	mAdc
Class A amp:	2,000	6.0^{+5}_{-10} %	330	-100	---	35	---	---	---
Class C Teleg:	2,000	6.0^{+5}_{-10} %	320	-100	---	35	---	---	15
Class C Teleph:	2,000	6.0^{+5}_{-10} %	260	-100	---	33	---	---	15
Anode pulse: (see note 2)	3,400	6.0^{+5}_{-10} %	---	-100	1,750	3.0	3.0	-100	1.3
TEST CONDITIONS:	---	6.0	200	---	---	---	---	---	---

ABSOLUTE RATINGS: MAXIMUM

Parameter:	ic	Pp	Pi	Dr	Ehk	Rk	tk	Mod	Seal Temp	Barometric pressure reduced
Unit:	a	W (see note 1)	W	---	V	Ohms	sec (min)	---	°C	mm Hg
Class A amp:	---	7.0	---	---	±90	---	60	---	175	55
Class C Teleg:	---	7.0	11.0	---	±90	---	60	---	175	55
Class C Teleph:	---	5.0	8.5	---	±90	---	60	Anode	175	55
Anode pulse: (see note 2)	1.3	6.0	---	0.001	±90	---	60	Anode	175	---
TEST CONDITIONS:	---	---	---	---	---	100	180	---	---	---

GENERAL:

Qualification - Required

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (see note 3)							
1266	Total grid current	Rk = 0; Ec = -2.5 Vdc	0.65	II	Ic	---	-1.0	µAdc
1256	Electrode current (1) (anode)		0.65	II	Ib(1)	16	34	mAdc
1256	Electrode current (2) (anode)	Rk = 0; Ec = -20 Vdc	0.65	II	Ib(2)	---	55	µAdc
1306	Transconductance		0.65	II	Sm	4,800	7,200	µmhos
1231	Pulse emission	eb = ec = 150 V max; prr = 1,000; tp = 0.9 to 1.1 µsec	0.65	II	ib	2.75	---	a
1236	Power oscillation (1)	eb = 1,750 V; Rg/Ib = 3.0 mA max; F = 3,300 ±100 MHz (see note 4)	0.65	II	po(1)	750	---	w
	Quality conformance inspection, part 2							
1301	Heater current		6.5	S3	If	260	300	mA
1211	Insulation of electrodes		6.5	S3	R	25	---	Meg
1236	Power oscillation (2) (CW)	Ef = 6.0; Eb = 350 V; Rk/Ib = 33 mA; F = 500 ±15 MHz	6.5	S3	Po(2)	4.5	---	W
1336	Heater-cathode leakage	Rk = 0; Ehk = +100 Rk = 0; Ehk = -100	6.5	S3	+Ihk -Ihk	--- ---	500 200	µAdc µAdc
1231	Emission	Rk = 0; Eb = Ec/Ik = 30 mA; Ef = 5.4 V	6.5	S3	Es	---	14	V
1331	Direct-interelectrode capacitance	See note 5	6.5	S3	Cgp Cgk	1.30 2.05	1.80 2.95	pF pF
1031	Low-frequency vibration(1)	Rk = 0; Ec = -2.5 Vdc; Rp = 10,000	6.5	S3	Ep	---	100	mVac

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3					
---	Life test (1) provisions (pulse)	Group B; F = 3,300 ±100 MHz; t = 100 hours (see notes 4 and 6)	---	---	---	---
---	Life test (1) end point	Power oscillation (1) (pulse)	po(1)	800	---	W
---	Life test (2) provisions (CW)	Group C; F = 500 ±15 MHz; t = 500 hours (see note 6)	---	---	---	---
---	Life test (2) end point	Power oscillation (2) (CW)	Po(2)	3.5	---	W
	Periodic-check tests					
1002	Barometric pressure, reduced	Pressure = 55 ±5 mm Hg; voltage = 500 Vac (see notes 7 and 8)	---	---	---	---
---	Seal fracture	No voltages (see notes 7 and 9)	---	50	---	lbs
---	Torque	No voltages (see notes 7 and 10)	---	15	---	lb-in
---	Lead bend	No voltages (see notes 7 and 11)	---	4	---	arcs
---	Low-frequency vibration (2)	No voltages; t = 180 sec (see notes 7 and 12)	---	---	---	---
1331	Direct-interelectrode capacitance	See notes 5 and 7	Cpk	---	0.07	pF
---	Frequency drift	See notes 4, 7, and 13	ΔF	---	3	MHz

NOTES:

1. If anode dissipation (Pp) exceeds 2.5 watts, special precautions must be taken to cool the anode externally in order to maintain the anode seal temperature below the specified maximum rating.
2. Total "ON" time shall not be more than 5.0 μsec in any 5,000 μsec interval.
3. All tests listed under quality conformance inspection, part 1, are to be performed at the conclusion of the holding period.
4. The tubes are to be tested in a reentrant oscillator cavity. Ic = 1.3 mA maximum; tp = 1.0 ±0.1 μsec; prr = 1,000 pps; RK = 0; pulse rise time = 0.1 μsec maximum; pulse fall time = 0.2 μsec maximum. Since the tube draws a certain amount of contact potential current (Ib and Ic) with Ef only, this current must be subtracted from the operating Ib and Ic readings in order to obtain the actual ib and ic currents during oscillation.
5. Capacitance shall be measured in adaptor as specified on drawing 155-JAN, or equivalent.
6. Cycle continuous.

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NOTES: -Continued

7. Tubes for periodic-check tests shall be selected at random at intervals so that when the tube is in production, at the time any lot is offered for acceptance, not less than six tubes will have been checked in the last six calendar months. Should the tubes fail a periodic-check test, that test will become a Quality conformance inspection, part 2 test (6.5 percent AQL, inspection level S3) until three successive lots pass. Thereafter, the test will again become a periodic-check test.

8. The specified voltage is to be applied between the anode and the grid plus cathode.

9. The tube shall be supported at the cathode and anode sleeves at points 15/16 ± 1/64 inch apart with the grid disc centered. The force shall be applied to the grid disc, without shock, perpendicular to the axis of the tube. The tube shall not fracture. This is a destructive test. Electrical rejects may be used.

10. The cathode terminal shall be rigidly clamped and a torque of 15 lb-in shall be applied gradually about the tube axis to the anode terminal. The tube shall not fracture. This is a destructive test. Electrical rejects may be used.

11. Each heater lead shall be subjected to a 16 ± 1 oz pull. Each lead must withstand the specified number of 90° arcs at the tube base. (An arc is described as the movement of the lead away from the tube axis and back to normal.) A lead fails at the time it breaks from the tube. Both leads are to be subjected to this test. This is a destructive test. Electrical rejects may be used.

12. The tube shall be vibrated in a plane perpendicular to the major axis. No voltages. Frequency = 40 to 60 Hz; t = 3 minutes; g = 10. At the conclusion of the vibration period, the tube shall meet the specified limits for+Ihk, Ib(2), Sm, and Vib(1).

13. Set for conditions of Power Oscillation(1). Reduce Ef to 5.4 volts; allow to reach temperature equilibrium and determine frequency. Increase Ef to 6.3 volts and allow to reach temperature equilibrium and determine frequency. The change in frequency shall be within the specified limit.

Custodians:
Army - EL
Navy - EC
Air Force - 11

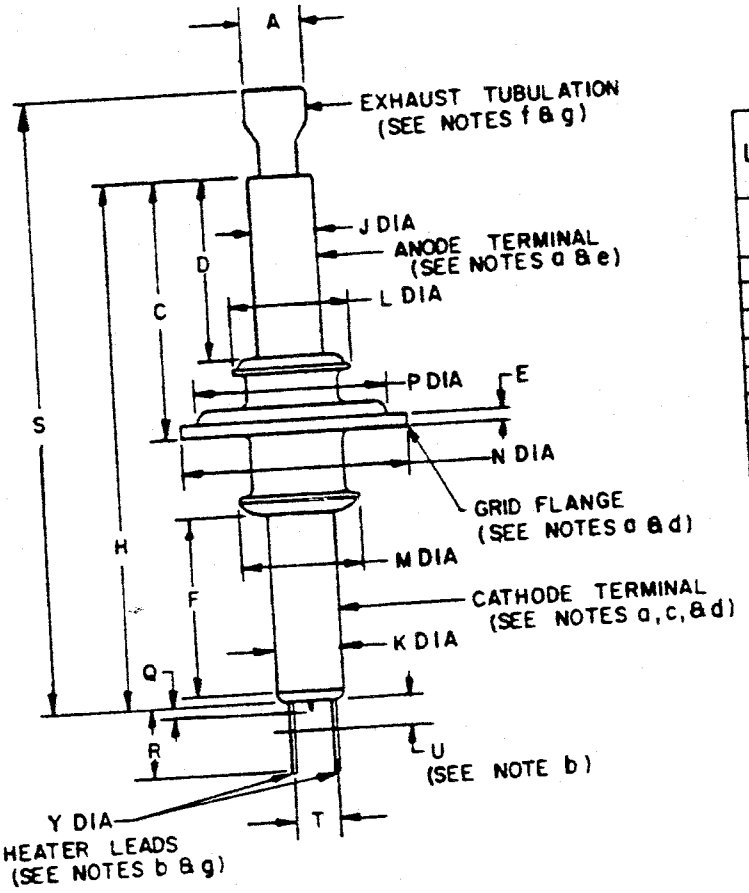
Preparing activity:
Navy - EC

(Project 5960-2240-1)

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 19

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3 (periodic check)		
A	- - -	.230 (5.84)
C	.845 (21.46)	.895 (22.73)
D	.600 (15.24)	- - -
E	.029 (.74)	.035 (.89)
F	.600 (15.24)	- - -
H	1.620 (41.15)	1.745 (44.32)
J	.245 (6.22)	.255 (6.47)
K	.245 (6.22)	.255 (6.47)
L	- - -	.400 (10.16)
M	- - -	.400 (10.16)
N	.807 (20.50)	.817 (20.75)
P	- - -	.687 (17.45)
Q	- - -	.040 (1.02)
R	.200 (5.08)	.240 (6.09)
S	- - -	2.057 (52.25)
T	.075 (1.91)	.155 (3.94)
U	- - -	.080 (2.03)
Y	.018 (.46)	.022 (.55)

NOTES:

a. Eccentricity of grid flange and of anode shall be .010 (.25 mm) maximum with respect to cathode terminal. Eccentricity of anode shall be gaged by chucking cathode terminal .050 (1.27 mm) minimum and .100 (2.54 mm) maximum from cathode glass seal and gaging 1/2 TIR, .080 (2.03 mm) from end of anode terminal. Eccentricity of grid flange shall be gaged by chucking cathode terminal as specified above and gaging 1/2 TIR of edge of grid flange. AQL shall be 6.5 percent and inspection level S3.

b. Leads shall be trimmed, and shall be tinned except as specified by dimension U. Untinned length of leads shall be measured from end of cathode terminal.

c. Glass from stem sealing shall not be present on cathode terminal surface. Glass button shall not extend radially beyond cathode terminal surface.

d. Allowable tilt of grid flange shall be .020 (.50 mm) maximum with respect to axis of cathode terminal and shall be determined by chucking cathode terminal, rotating tube through one complete revolution, and gaging total travel distance of flange parallel with axis of cathode terminal and .020 (.50 mm) from edge of flange. AQL shall be 6.5 percent and inspection level S3.

e. Diameter of anode terminal shall not be increased by solder or melted silver from exhaust tubulation.

f. Allowable burned area at exhaust tubulation end of anode terminal shall be .0025 square inch maximum. AQL shall be 6.5 percent and inspection level S3.

g. All external metal surfaces, except leads and exhaust tubulation, shall be silver or gold plated 50 msi minimum. Plating shall not be blistered and plated areas shall exhibit no exposed base metal.

FIGURE 1. Outline drawing of electron tube type 5893.

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